Enrico Mattia Salonia
Game Theory - Spring 2026

How to Study for Game Theory

These are some thoughts on how to study for the game theory class. They are a mix of my own experience and research on learning mathematics that I read in (Alcock, 2021) (which is a fun read). All the empirical claims in this note are documented by research collected in that book. Of course, none of this is compulsory: study in whatever way works for you. Also, do not "try" the exam, it is a bad idea. You may feel it is useful, but psychologically it licenses you to work less consistently. Moreover, I usually post past exams, so you can have a look at them even if you do not enrol.

As for studying, the best way to prepare for this class is to read the book consistently and carefully, attempt exercises from the first day, write definitions and solutions by hand on paper, and continuously ask yourself whether you actually understand what is going on. What I mean is perfectly captured by the following:

> Because logic is innate, the laws of logic that you learn should make sense—if you find yourself having to memorize one of the principles or laws of logic here, without feeling a mental "click" or comprehending why that law should work, then you will probably not be able to use that law of logic correctly and effectively in practice. So, please don't study this appendix the way you might cram before a final—that is going to be useless. Instead, put away your highlighter pen, and read and understand this appendix rather than merely studying it! (Tao, 2006, p. 307)

The paragraph is about logic, but works for game theory too. Let me elaborate.

First, studying a mathematical subject is different from studying other subjects. Most of us are not trained to "read" mathematics and engage deeply with its logic as we read. For this reason, studying should feel hard. If you are struggling, it means you are learning—not that you are doing something wrong. Conversely, there is no hope of gaining a deep understanding of the subject without struggling. This class is for beginners, so the approach is meant to be gentle, but it may still feel difficult.

There are two fronts to pay attention to: how you organise your study, and how you do it. There is no secret to the first: organise yourself, work consistently, space different topics out, and give things time to sink in. The second deserves more comment. You will not gain a deep understanding by passively attending lectures or skimming notes and the book. If you want to learn how to compute Nash equilibria in mixed strategies and you just skim your notes, you may be tempted to think, "Ah yes, I remember this"—but probably you don't. You have to pick up a pencil and write down how to compute Nash equilibria in mixed strategies. The same goes for every part of the class.

I suggest the following. You know what the exam consists of:

1. Read the book carefully. For each piece of notation, ask yourself, "Is it clear to me what this means?" For each definition, try to write it down on paper without looking at the book, and ask, "Do I understand what this definition means? Can I explain it to myself in English? Can I come up with a simple example?" For each new concept, ask, "How is this related to concepts I've already studied?"

2. Do as many exercises as you can, mindfully. Read the exercise and attempt it before looking at the solution. This will sometimes be painful. But if you push through, you will learn the material well. When reviewing exercises, don't just skim the solution and think, "Yes, I remember this"—chances are you don't.

3. Practice self-explanation. Self-explanation is a technique researchers use all the time (even when they don't know they're doing it). The idea is to explain to yourself what you are reading and doing. It forces you to notice when you don't understand something, which is uncomfortable but necessary. The questions in points 1 and 2 are meant to guide you in doing this. While self-explaining, don't fall into the trap of paraphrasing the text or telling yourself, "I got this." Actively extract and elaborate the ideas. You can find an example in (Alcock, 2021) if you are interested.

The exam consists of:

1. A definition. To get it right, you need to study and understand definitions. In the exam of April 2026, more than 90% of students got the definition of Nash equilibrium wrong—not "roughly wrong", but very wrong. Memorising definitions by heart is risky because there are many of them; understanding them is far safer.

2. Two true/false/uncertain questions, one on normal-form and one on extensive-form games. There is no shortcut here: to answer correctly, you need to understand the concepts of the class. These are not trick questions. As an example, in the exam of April 2026 many students asked me about the second true/false/uncertain question: "How can I know the players' strategies if the exercise doesn't give the payoffs?" You don't need the payoffs to list the strategies—but you do need to understand what strategies and payoffs are in order to see that.

3. Three exercises, one of which will be slightly harder. There is a relatively small set of "algorithms" we covered in class and that appear in the book's exercises. You need to do many exercises to learn how to apply them to new problems. The vast majority of students in the exam of April 2026 could not find strategies dominated by mixed strategies, nor compute Nash equilibria in mixed strategies. We did this many times in class, there are plenty of exercises on it, and there was even a very similar problem in the mock exam. But if you don't work through it yourself a few times, you won't be able to do it during the exam.

Lastly, have fun!

References

Alcock, L. (2021). *How to think about abstract algebra.* Oxford University Press.
Tao, T. (2006). *Analysis* (Vol. 1). Springer.